Ivar Cashin Eriksson

Curriculum Vitae

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Introduction

I have seven years' experience applying machine learning in healthcare, pharmaceutical analytics, and consulting, with training in Engineering Physics and Statistical Learning and Data Analytics at KTH.

My practical experience includes probabilistic ML, explainable AI, production ML, and clinically sensitive prediction problems. I am particularly interested in how uncertainty, robustness, fairness, and interpretability affect the safe use of machine learning in health-data settings.

Education

2015–2020 **Royal Institute of Technology, KTH – Engineering Physics**, GPA: 5.0 / 5.0
Master: Statistical Learning and Data Analytics, Bachelor: Engineering Physics. See thesis.
Relevant coursework: Probability Theory; Statistical Machine Learning; Computational Statistics.

2017–2022 **Stockholm University, SU - Economics**, *Supplementary Bachelor's Degree*

2019–2019 **Swiss Federal Institute of Technology, ETH - Applied Mathematics**, *Exchange*

Selected Work Experience

Researcher within Data Science @ RaySearch Laboratories

- Developed radiotherapy planning pipelines using deep learning, probabilistic ML, and optimisation.
- Built autoencoder-based features from segmented 3D CT scans and sparse Gaussian process models for uncertainty-aware, probabilistic dose prediction.
- Worked closely with clinicians to ensure model outputs were clinically relevant and interpretable.

Focus: clinical AI, probabilistic modelling, deep learning, embedding learning, optimisation, medical imaging

Read more...

Principal Data Scientist @ Valcon

- Led data science projects across pharma, energy, IoT, and manufacturing.
- Implemented an end-to-end production pipeline for a real-time deep neural network and led explainable AI training.

Focus: production ML, explainable AI, model evaluation, stakeholder communication

Read more...

Full-Stack Data Scientist @ Signum Life Science

- Maintain and develop a pharmaceutical price and demand forecasting platform with reproducible model evaluation workflows.
- Own and extend a next-best-action system, including explainable AI features and embedding-based matching workflows.

Focus: forecasting, explainable AI, ownership of production ML systems

Read more...

Skills and Other

Technology	Python, SQL, PyTorch, TensorFlow, Scikit-learn, SHAP, XGBoost, FastAPI, Databricks.
Methods	Uncertainty-aware probabilistic modelling, deep learning, model evaluation, explainable AI, sparse Gaussian processes.
Language	*Native*: English, Swedish. *Intermediate*: Danish.

Other	Valcon People's Choice Award for exemplary work, team spirit, and curiosity; Mensa Sweden member.

Detailed Work Experience

2024–Present **Full-Stack Data Scientist and Project Manager**, *Signum Life Science*, Copenhagen

At Signum, my current work is on an end-to-end forecasting platform for pharmaceutical price and demand forecasting. The work builds on a pharmaceutical price forecasting model that helps clients navigate Denmark's fortnightly blind auctions. After stabilising the XGBoost-based system, I am now driving efforts toward a more robust and modern implementation made to support thousands of production forecasting models, with emphasis on reproducible evaluation, robust performance across heterogeneous product groups, and clear communication of model limitations.

Earlier in the role, I owned and extended Kwarts, a next-best-action engine for pharmaceutical sales. Originally built externally, the system was extended with features such as explainable AI in collaboration with UX designers, and supported commercialisation by aiding our sales representatives. Kwarts adapts to varied customer data maturity, from rule-based logic to deep learning, and is deployed in AWS for scalability.

Beyond model development, I organised an internal week-long hackathon that included up to 40 participants.

2022–2024 **Principal Data Scientist**, *Valcon*, Copenhagen

Led data science projects across pharma, energy, IoT, and manufacturing. Mentored junior colleagues in software architecture and data science best practices, led explainable AI and Git trainings, and regularly presented complex technical topics to non-technical stakeholders.

One project involved training a real-time deep neural network with a custom training pipeline and modular architecture. The training data consisted of natural-language inputs and high-dimensional multi-class labels, which made for a challenging prediction problem. I was the lead data scientist and architect of the solution and responsible for implementing the end-to-end production pipeline. An important part of the work involved communicating model behaviour and limitations to non-technical stakeholders.

I was awarded the "Valcon People's Choice Award" for exemplary work, leadership, team spirit, and scientific curiosity.

2021–2022 **Data Scientist**, *Valtech*, Copenhagen

Delivered ML solutions in e-commerce. Built time series forecasts using Facebook Prophet, and applied clustering to behavioural data to support personalisation. Worked closely with business stakeholders to ensure actionable outputs.

2019–2021 **Researcher within Data Science**, *RaySearch Laboratories*, Stockholm

At RaySearch Laboratories, I worked on automating radiotherapy planning using a combination of deep learning, probabilistic modelling, and multi-objective optimisation. I developed a pipeline that extracted latent anatomical features from segmented 3D CT scans using autoencoders, and used these representations to identify similar patients via a learned similarity metric. Dose-volume histograms and spatial dose distributions were then predicted using sparse Gaussian processes, providing an uncertainty-aware framework for dose planning conditioned on patient geometry.

Alongside this, I redesigned some of RaySearch's lexicographic optimisation algorithms to better reflect clinical decision hierarchies, particularly in palliative care where trade-offs between tumour control and organ sparing are nuanced. I collaborated closely with medical physicists and clinicians throughout, ensuring the models aligned with oncological standards and practical workflow requirements.

Selected Technical Project

2025–2025 **Solo Project**, *Project Iris*

In this personal project I built a PoC end-to-end computer vision system for making product images on e-commerce sites interactive. The pipeline detects individual items using a YOLOv8 model, and embeds them semantically using OpenCLIP. The embeddings are then compared to support intelligent product linking across all images and products in a web store.

The system includes a full-stack setup with a FastAPI backend, MongoDB for persistence, a Qdrant index for vector search, and a custom-built Chromium plug-in that overlays clickable links directly onto web shop imagery in real time. The project also includes a dynamic scraping and indexing pipeline for ingesting and structuring product data from arbitrary e-commerce sites.